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SmartHoneypot: Honeypot System for Lightweight Cyber Threat Detection and Behavior Analysis

Dharshan J Y, Arjun R Amarnath, Gadamsetty Siva Adithya, Anitha V, Kumaran U.
Department of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
bl.en.u4cse22219@bl.students.amrita.edu, bl.en.u4cse22207@bl.students.amrita.edu,
bl.en.u4cse22224@bl.students.amrita.edu, bl.sc.r4cse24011@bl.students.amrita.edu, u_kumaran@blr.amrita.edu

***Abstract*—The identification and classification of cyberattacks were achieved by deploying Cowrie as a medium-interaction SSH/Telnet honeypot that mimicked an exploited corporate server environment. In order to develop realistic attacker patterns, we ran attack scenarios such as brute-force login tests, mock malware uploads and ransomware-styled operations. Cowrie monitored steps of these events, logging in, using commands, and file-related activities. Extraction of behavioural indicators from the logs was placed into a structured database containing session duration, command usage, and login behaviours. By applying the data gathered, a random forest classifier was developed for the purpose of classifying sessions based upon known attack patterns. After its training, the system was able to automatically label incoming sessions and identify potential types of attacks by profiling user behaviour. To increase awareness and response capabilities, each attack category was linked to relevant security guidance and shielding tactics, including advocating for strong password practices and strengthening intrusion notification and access privilege control. Moreover, an interpretable format was given by appending visualisation tools that emphasise session metrics and predictive models. By combining honeypot data with machine learning, this process is used to point out the capability of using this technique to discover an attacker's activities, used to systematically classify different types of attacks, and eventually, offer meaningful insights about cybersecurity threats.**

***Index Terms*—brute force, malware, ransomware, threats, security**

I. Introduction

With increasing attacks and their complexity and frequency, many organizations now find that their installed security measures are not sufficient to detect and interpret the techniques of an attacker. Honeypots are able to help researchers learn about attacker tactics by allowing them to create deceptive environments to attract cybercriminals. Subjecting attackers to a simulated environment helps to reveal some of the main points about the ways they try to infiltrate systems, deliver malicious payloads, and function in the long run. Cowrie stands out among the garden of available honeypot products as a medium-interaction SSH/Telnet honeypot that mimics a shell, logs attempts to log in, and logs the commands of the attackers.

Cowrie was used for this project as a corporate database server and for monitoring a variety of attack behaviors, including brute-force logins, malware uploads, and even ransomware simulations. All recorded interactions were presented in JSON format and gave important data for further analysis. Having structured the logs and equipped with features that can reflect the behavior of a given user, a Random Forest classifier was trained to automatically assign to each session—based on those observed actions—a certain category. This approach is based on machine learning to enable real-time attack detection and facilitates sophisticated categorization that comes in handy for threat response and forensics analysis.

In addition to detection, the system provides actionable security recommendations and vents for a type of attack detected. For instance, the system suggests that strong password rules should be required, closely monitor file sharing activity, and limit the use of remote shells. Visual dashboards allow understanding the data better, with the help of categories of attack types, session metrics and detection results. With incorporation of honeypot data, behavioral analysis, machine learning and awareness generation, this system offers a practical and learning tool for enhancing cybersecurity preparedness.

The rest of the paper is structured as follows: Section II describes the previous works done on this topic. Section III outlines how the research was conducted, detailing the approach, procedures, and techniques used to achieve the study's objectives. Section IV presents the results along with discussion as a critical analysis of its performance, and insight obtained from it. Section V concludes the present study, as it summarises the key findings and gives way for future directions.

II. Review of Related works

Mummaneni et al. [1] studied the tactics used in adversarial evasion attacks that exploit the structure of PE files to evade detection by MalConv. The completeness of padding and slack space complemented each other, resulting in a significant improvement of evasion efficiency. Reading from Nguyen et al. [2], static analysis augmented by Random Forest and 1D-CNN proved to be of great assistance in classifying different families of Android malware. Although classification successes were high, CNN-based models did not completely avoid the cases of false positives.

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Control Flow Graphs were used together with algorithmic, formal, and machine learning techniques to detect malware by Mircea-George and Pungila [3]. This technique is found useful, but uncontrolled consumption of resources and absence of scalability require optimising this technology for large scale application. Kumar et al. [4] proposed a hybrid malware classification model combined with the usage of the Cuckoo Search Algorithm for feature selection and LSTM networks. Their joint system performed better than usual methods, but their real-world, dynamic malware settings were not part of their testing.

Ferdous et al. [5] reviewed the evolution of malware as well as presented multiple multi-layered methods for detection, paying particular attention to AI-orientated approaches such as CNNs and federated learning. Despite the variety of the methodologies covered by the study, it failed to consider the difficulties encountered in implementing the system. Samsoni et al. [6] investigated keyloggers and their attack mechanism, detailing the attack mechanisms by both kernel-based and API-based. While it highlighted the inherent limitations of the traditional tools in such identification, it provided countermeasures.

Jimmy [7] reviewed the development of ransomware and classified the defence mechanisms as static, dynamic or a combination of AI models. Although the work highlighted the multi-layered detection, it exposed real-world shortcomings and open issues related to the management of zero-day vulnerabilities. On the UNSW-NB15 dataset, Azeem et al. [8] considered several ML classifiers with a focus on entropy-based feature characteristics in particular. Random Forest and Extra Trees performed almost perfectly with accuracy, but the study failed to evaluate the deep learning technique and failed to validate findings using another dataset.

E.V.C. et al. [9] have done a detailed analysis of logger detection that includes phishing, anti-hook techniques, as well as memory forensics. Detection of keyloggers in user space was contemplated, but kernel-level defences were not considered in the research.

Bejo et al. [10] created a motherly Python-based keylogger that records keystrokes and usage of mobile apps without falling foul of many virus scan engines. Despite its intention to go unnoticed, its high requirements on the CPU undermined performance, forcing the need to move to more efficient solutions for monitoring. By using SVM classifiers and feature selection methods, Balakrishnan and R. P. N [11] were able to detect keyloggers based on keystroke sequences of the user. The model did well, but it struggled to identify advanced or obfuscated keylogger variants.

Srinivasan [12] developed a cross-platform keylogger detector framework using XGBoost, LightGBM and CatBoost complemented with careful feature engineering. With good performance, the dataset did not capture the massive variety of the keylogger types and methods of evasion. A ransomware detection framework was proposed by Khaliq et al. [13], where Kaspersky tools are combined with VM-based proxy firewalls to quarantine and observe suspicious actions. Although the approach provides hope, it may be susceptible to new, zero-day, or well-obfuscated attacks. Bhat et al. [14] proposed a cybersecurity testbed for keylogger simulation and visualisation on the basis of Python and Kaggle data sets. The model has educational applications but lacks strong systems for handling several types of input events and accepting a range of data streams.

Edie et al. [15] proposed updating threat playbooks to support APT attribution by analysis of TTPs and STIX similarity metrics based on data from MITRE ATT&CK. The system comes with outstanding attribution results, yet its success is dependent on regular threat database updates. Thakar et al. [16] have done a review of cyber threats supported by darknets and developments in terms of cyber security, such as the use of AI in IDS and incident response. Though comprehensive, the review lacks evidence from experiments or practical steps to implement the techniques proposed.

Kumaran et al. [17] present HoneySQLGuard, an intelligent system for defence against SQL injection attacks, which integrates honeypot-type tables into database architectures and utilises anomaly detection for real-time protection. It uses deception-based tables and machine learning to identify an attack pattern and dynamically adapt defence strategies.

A work by Kumaran et al. [18] reflects a well-rounded set of combined analyses, which includes malware behaviour analysis, trends in phishing attacks, and analysis in sentiment related to cybersecurity events. It assesses malware evasion techniques, phishing campaigns on trusted platforms, and media sentiments on data breaches in order to provide the bigger picture of cyber threats.

Niharika et al. [19] come up with a honeypot-based intrusion detection system that identifies unauthorised logins, captures attacker images through web cameras, logs IPs and attempted credentials and sends alerts to admins. The system is a combination of deception and real-time monitoring combining to form an active user-level defence layer.

Detection and prevention strategies for DDoS attacks against cloud environments are reviewed by B. S. and Kartheek [20], among other things, in honeypot configurations. They suggest a modular cloud honeypot structure that should accept, record, alert, and redirect malicious traffic based on the techniques of virtualisation and SDN.

III. Methodology

A lightweight SSH honeypot Cowrie is deployed to simulate an insecure server environment for the test purposes. To lure attackers, Cowrie is set up to execute itself on an port where it will watch for and log SSH login attempts. It sets up a virtual file system and command line and maintains an accurate record of all the actions of the attackers, whether login, command or upload of files. Interaction is logged into structured JSON logs, named cowrie.json, with key fields being event ID, source IP, input, timestamp, and session ID. Analysis of intruders is carried out by partitioning and sorting logs on separate user sessions. By using this data, we are able to recognise early warnings of possible security problems. Such logs are analysed to derive statistics on failed logins,

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command usage, and indicators of possibly malicious action. From this process, raw attacker activities are extracted into useful features, labelled and ingested into a machine learning model to enable predictive classification and awareness.

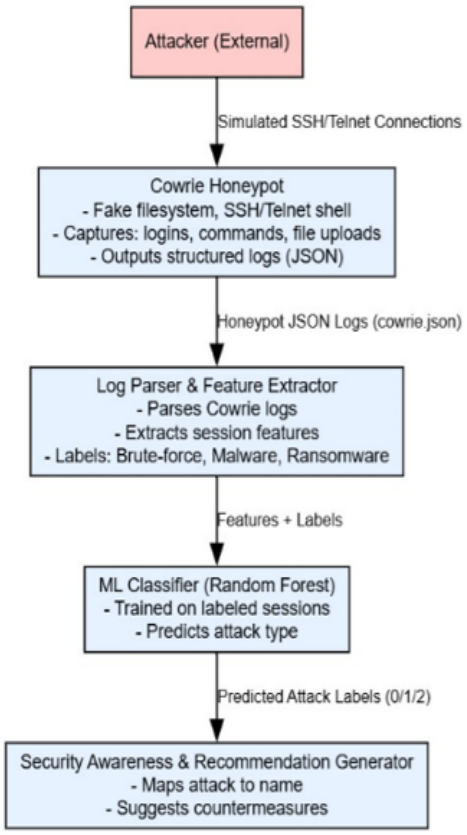


Fig. 1. System Architecture

The system architecture is shown in Fig. 1. Three different sets of attack scenarios were used to attack the Cowrie honeypot in order to simulate real-world threats. First, the Hydra tool was used to execute brute-force attacks, in which different passwords were used for trying to access at high speed. Several cowrie.login.failed were logged by cowrie from these attempts. Secondly, in order to simulate malware behaviour, the honeypot uploaded and pretended to run shell scripts. The third one is ransomware simulations, which included the creation of threatening messages and encrypting files through OpenSSL commands. The resulting commands and file-related activities were written in the Cowrie log entries. Each simulated attack left a specific pattern of the Cowrie event records. The analysis and matching of these patterns among different sessions will provide automated identification of the attack type and how each behaviour manifests itself in logs.

Structured log data obtained from Cowrie sessions are used to train a random forest classifier, which is a multiclass supervised learning method based on ensembles of decision trees. Random Forest upwards the accuracy and prevents the occurrence of overfitting by the formation of an ensemble of trees that work on different subsets of the training data. The decision is because it works well with high-dimensional, noisy data types and works well with cluttered data, which is observed during attacker sessions. The predictors include the metrics such as the number of failed login attempts, the occurrence of download commands, total commands to run, and detection of keywords such as 'ransom', 'wget' or 'chmod'. After training the model with labelled sessions (0 for brute-force, 1 for malware, and 2 for ransomware), the same model is applied to unseen new sessions for the evaluation. Random Forest then, with the use of its predictive model, analyses the log data to determine the most probable attack pattern, which hence allows for appropriate countermeasures and awareness recommendations to be exercised.

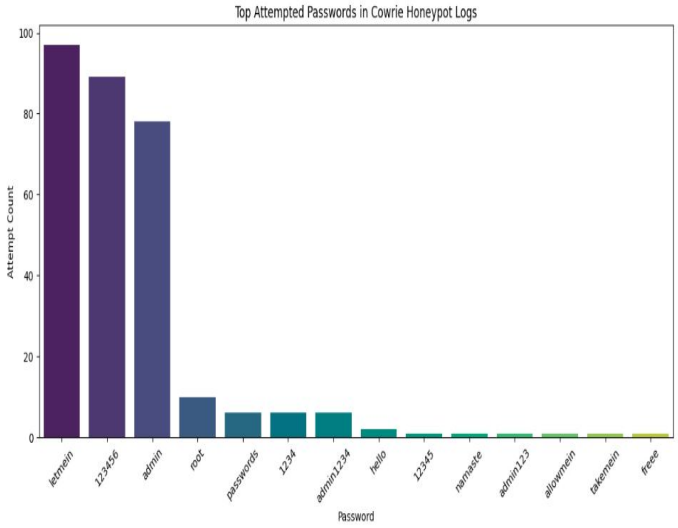


Fig. 2. Top passwords used

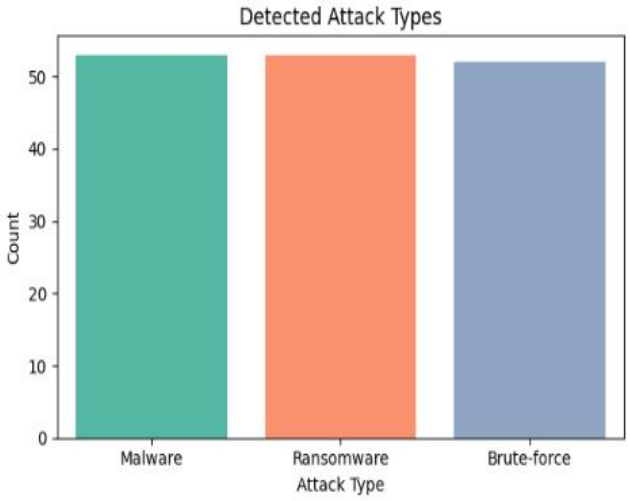


Fig. 3. Different Attacks used

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Classification Report:

	precision	recall	f1-score	support
Brute-force	1.00	1.00	1.00	14
Malware	0.94	0.94	0.94	16
Ransomware	0.94	0.94	0.94	18
accuracy			0.96	48
macro avg	0.96	0.96	0.96	48
weighted avg	0.96	0.96	0.96	48

Fig. 4. Performance metrics

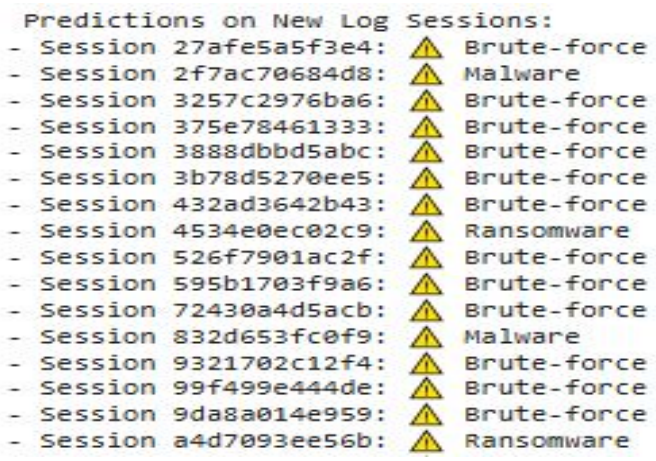


Fig. 5. Predictions

IV. Results and Discussions

With the help of Cowrie-log-derived features, the Random Forest model is able to classify attack sessions, as seen in Fig. 5 into one of the attacks, such as brute force, malware, or ransomware. An effective impact on prediction accuracy was noticed for features such as failed_logins, command_count and the use of keywords like 'wget' and 'chmod', which helped to classify the attacks easily. Finally, the model demonstrated noticeable transformability regarding new log files, successfully distinguishing sessions with the use of behavioural analysis. With these applications of the log parsing and feature extraction techniques, It transformed the raw honeypot logs into usable security insights that formed a strong dataset for threat analysis and classification.

Different attack categories, as shown in Fig. 3, were well separated by graphs generated using Seaborn and Matplotlib. For brute-force attacks, there were many login attempts but few commands in comparison to the malware simulations, where many commands overwhelmed and file activities were dominant. Ransomware activity was commonly tracked with unique keywords and echoes of extortion messages. These findings demonstrate that the combination of honeypot interaction logs and machine learning techniques would lead to an efficient, simple intrusion detection and awareness mechanism, which will secure real systems.

V. Conclusion

Cowrie-based honeypot systems successfully emulate real cyberattacks, like brute force breaches where they show the top passwords used, as shown in Fig. 2, malware dissemination, and ransomware operations. Precise attack logs obtained by the system made it possible to develop valuable datasets that could be used to analyse security incidents. Using the collected logs, a machine learning pipeline based on the Random Forest classifier as seen in Fig. 4 was built, allowing for accurate classification of distinct attack types based on observable patterns. Intelligent attack detection techniques and deception strategies are immensely beneficial in expanding threat identification capabilities and revealing essential facts about attackers' purposes. The amalgamation of ML and honeypots in this study shows a promising approach towards improving cybersecurity defence systems, strengthening the network protection, detecting anomalous activity, and providing advanced countermeasure direction.

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